THE ALTERNATING CECH COMPLEX AS THE COLIMIT OF DUAL KOSZUL COMPLEXES

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Allowed standard prerequisites (editorial).

- The alternating Cech complex is the subcomplex of the usual Cech complex given by alternating cochains, whose p-cochains are alternating functions on (p+1)-tuples of indices with values in the sections over the corresponding intersection. used at proof lines 2142-2152. Fixes the target complex in which the image of Psi is described.
- The transition maps of the inverse system of Koszul complexes send the basis wedge [i_0,...,i_p] to the product of the functions times itself, matching the denominators of the formula. used at proof lines 2158-2166. Ensures Psi is compatible with the colimit system.
- Localized modules are described as colimits over powers of the inverting elements. invoked at proof lines 2167-2171. Shows Psi is an isomorphism of cochain modules in each degree in the colimit.
- The differentials on the Koszul complex used in the system are the negatives of the standard ones. used at proof lines 2178-2210. Accounts for the sign in the differential compatibility computation.

Source reference. Original remark 08CZ: sources/stacks-perfect/pinned-original-source.tex, source lines 2116–2116. Referenced at statement line 2128 for the setup; the construction of Psi within this situation is inside the counted span.

Source reference. Original section 01FG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2143-2145 for the definition of the target complex; the explicit identification is inside the counted span.

Source reference. Original equation 08CY: sources/stacks-perfect/pinned-original-source.tex, source lines 2107–2107. Referenced at proof lines 2159-2163 for the transition maps; the compatibility check with them is inside the counted span.

Source reference. Original lemma 00CR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2167-2171 for the degree-wise surjectivity and injectivity of Psi; the differential computation finishing the proof is inside the counted span.

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Lemma 0.2. *In Situation 08CZ. Let M be an A-module and denote $\mathcal{F}$ the associated $\mathcal{O}_X$-module. Then there is a canonical isomorphism of complexes*

$$\Psi : \operatorname{colim}_e \operatorname{Hom}_A(I^\bullet(f_1^e, \ldots, f_r^e), M) \longrightarrow \check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F})$$

functorial in M where the differentials on the Hom*-complex are the contragredients of the differentials on $I^\bullet(f_1^e, \ldots, f_r^e)$.*

Proof. Recall that the alternating Čech complex is the subcomplex of the usual Čech complex given by alternating cochains, see Cohomology, Section 01FG. As usual we view a p-cochain in $\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F})$ as an alternating function s on $\{1, \ldots, r\}^{p+1}$ whose value $s_{i_0 \ldots i_p}$ at $(i_0, \ldots, i_p)$ lies in $M_{f_{i_0} \ldots f_{i_p}} = \mathcal{F}(U_{i_0 \ldots i_p})$. On the other hand, a p-cochain t in $\operatorname{Hom}^\bullet(I^\bullet(f_1^e, \ldots, f_r^e), M)$ is a map $t : \wedge^{p+1}(A^{\oplus r}) \to M$. Write $[i] \in A^{\oplus r}$ for the ith basis element and write

$$[i_0, \ldots, i_p] = [i_0] \wedge \ldots \wedge [i_p] \in \wedge^{p+1}(A^{\oplus r})$$

For t as above we set

$$\Psi(t)_{i_0 \ldots i_p} = (-1)^p \frac{t([i_0, \ldots, i_p])}{f_{i_0}^e \ldots f_{i_p}^e}$$

It is clear that $\Psi(t)$ is an alternating cochain. The rule above is compatible with the transition maps of the system as the transition map

$$I^\bullet(f_1^e, \ldots, f_r^e) \leftarrow I^\bullet(f_1^{e+1}, \ldots, f_r^{e+1}),$$

of (08CY) sends $[i_0, \ldots, i_p]$ to $f_{i_0} \ldots f_{i_p}[i_0, \ldots, i_p]$. It is clear from the description of the localizations $M_{f_{i_0} \ldots f_{i_p}}$ in Algebra, Lemma 00CR that the rule Ψ defines an isomorphism of cochain modules in degree p in the colimit. To finish the proof we have to show that the map is compatible with differentials. To see this, for t as above we compute

$$\begin{aligned} d(\Psi(t))_{i_0 \ldots i_{p+1}} &= \sum\nolimits_{j=0}^{p+1} (-1)^j \Psi(t)_{i_0 \ldots \hat{i}_j \ldots i_{p+1}} \\ &= (-1)^p \sum\nolimits_{j=0}^{p+1} (-1)^j t([i_0 \ldots \hat{i}_j \ldots i_{p+1}])(f_{i_0} \ldots \hat{f}_{i_j} \ldots f_{i_p})^{-e} \end{aligned}$$

Recall that the differentials on $I^\bullet(f_1^e, \ldots, f_r^e)$ are the negative of the differentials on $K^\bullet(f_1, \ldots, f_r)$. Thus

$$\begin{aligned} \Psi(d(t))_{i_0 \ldots i_{p+1}} &= (-1)^{p+1} d(t)([i_0, \ldots, i_{p+1}])(f_{i_0} \ldots f_{i_{p+1}})^{-e} \\ &= (-1)^{p+1} t(d([i_0, \ldots, i_{p+1}]))(f_{i_0} \ldots f_{i_{p+1}})^{-e} \\ &= (-1)^{p+1} t(-\sum\nolimits_{j=0}^{p+1} (-1)^j f_{i_j}^e [i_0, \ldots, \hat{i}_j, \ldots i_{p+1}])(f_{i_0} \ldots f_{i_{p+1}})^{-e} \\ &= -(-1)^{p+1} \sum\nolimits_{j=0}^{p+1} (-1)^j t([i_0, \ldots, \hat{i}_j, \ldots i_{p+1}])(f_{i_0} \ldots \hat{f}_{i_j} \ldots f_{i_p})^{-e} \end{aligned}$$

The two formulas agree concluding the proof. □